LRS31 Asset Pricing and Portfolio Theory

Home Assignment 1

Use RStudio to complete all numerical calculations, data processing, statistical tests, portfolio optimisation, and plots in R. Show analytical derivations where requested. Submit your R script (`.R`) or R Markdown source and relevant R output. Your code must reproduce your reported results.

1. **Limits to diversification.** Consider an equally weighted portfolio of N securities. First assume that returns are independent and identically distributed, with finite variance. Prove that portfolio variance approaches zero as N increases.

   Next, assume that the average individual variance converges to a finite limit and the average covariance between distinct securities converges to a strictly positive limit. Prove that portfolio variance does not approach zero.

   a. Managed to prove both parts.
   b. Managed to prove only either part.

2. You have a portfolio consisting of one company, MyOwnStartUp, in which you have invested x euros. Due to an ongoing special situation, the company is in danger of bankruptcy. If bankruptcy occurs, you lose 80% of your investment; 20% can be recovered by selling the company's assets. You estimate the probability of bankruptcy as $p = 10\%$. Otherwise, you expect a return of $+15\%$. Assuming the utility function

$$U(R) = \mathbb{E}[R] - \frac{1}{2}\lambda \operatorname{Var}(R),$$

   calculate the expected utility score for risk aversion $\lambda = 2$.

   a. $U < -2$
   b. $-2 < U < -1$
   c. $-1 < U < 0$
   d. $0 < U$

3. **Q2 continued.** Assume $x = 100\,000$ euros. You are offered insurance costing $1\,000$ euros that reimburses the investment lost in the event of bankruptcy. You are willing to take the insurance, but would not take it if it cost one cent more. Calculate your risk aversion λ.

   a. $\lambda < 0.5$
   b. $0.5 < \lambda < 1$
   c. $1 < \lambda < 2$
   d. $2 < \lambda$

4. Download `HA2_data_adj.xlsx` from Moodle. On `Sheet1`, columns C–I contain a 121×7 matrix of monthly total return index values for assets/market indices. In R, calculate continuously compounded returns to obtain the 120×7 return matrix $\mathbf{R}$. Calculate $\mathbf{R}^\top\mathbf{R}$ and sum its diagonal entries. What is the sum? *R hint:* use `crossprod()`, `diag()`, and `sum()`.

   a. SUM < 0

b. $0 < \text{SUM} < 5$

c. $5 < \text{SUM} < 10$

d. $10 < \text{SUM}$

5. Calculate the mean continuously compounded return and volatility of each of the seven assets in Q4 using R. Sum all 14 values. What is the sum?

a. $\text{SUM} < 0$

b. $0 < \text{SUM} < 0.5$

c. $0.5 < \text{SUM} < 1$

d. $1 < \text{SUM}$

6. Assume the utility function

$$U(W) = W - \frac{\lambda}{2}W^2, \qquad \lambda > 0.$$

Restrict attention to the region where utility is increasing. Derive absolute risk aversion (ARA). Then prove that this quadratic utility function exhibits increasing absolute risk aversion (IARA).

a. Managed to do both.

b. Managed to derive only ARA.

7. Download the monthly Fama/French three-factor returns and risk-free return from Kenneth French's Data Library. The four series are the market excess return, size factor return, value factor return, and risk-free return. They are simple monthly returns expressed in percent: divide all values by 100 to obtain decimals. Remove the annual observations at the end of the file. Using R and all four monthly series, calculate each series' mean, standard deviation, and variance, and calculate the variance–covariance matrix. What is the sum of all 16 entries of this matrix?

a. $\text{SUM} < 0$

b. $0 < \text{SUM} < 0.5$

c. $0.5 < \text{SUM} < 1$

d. $1 < \text{SUM}$